

Non-linear association joint models for clinical studies

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WST795 Research Report

Submitted in partial fulfillment of the degree BSc(Hons) Mathematical Statistics

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Abstract

Short summary of the research proposed. This should be a two to three paragraphs long and should fully describe the content and contributions of the research report.

Joint models in a clinical context aim to simultaneously model a patients longitudinal biomarker data and their Time-To-Event (TTE) survival data. For the sake of statistical, mathematical, theoretical, and practical simplicity, a linear linkage function between the two processes is assumed. However, this may easily fail to capture the complex and often non-linear processes and mechanisms that characterize pharmacokinetics and pharmacodynamics.

This study aims to investigate the effects of using a non-linear association function in a joint model within a clinical context; advantages, disadvantages, and general efficacy will be investigated and discussed.

Declaration

I, *Sable Walters-Somers*, declare that this research report, submitted in partial fulfillment of the degree *BSc(Hons) Mathematical Statistics*, at the University of Pretoria, is my own work and has not been previously submitted at this or any other tertiary institution.

Sable Walters-Somers

Janet van Niekerk

Date

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1 Introduction

YOUR 10 PAGE COUNT STARTS HERE

The introduction should provide a detailed description of the topic and the aim of the research report. In addition the literature review should intertwine with this. It is important to always reference where needed. All work from somewhere else requires a reference ?.

QUESTIONS/POINTS TO COVER:

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Inline equation: $x = y$

Display equation:

$$x = y$$

Numbered display equation:

$$x = y \tag{1}$$

Numbered equation:

$$x = y + 1 \tag{2}$$

Equation array:

$$\begin{aligned} x &= y + 2 + 3 \\ &= y + 5. \end{aligned}$$

2 Background Theory

In Figure ?? and in Figure ?? we see the UP logo.



Figure 1: A graphic of the UP logo

- What should be in a caption?

Figure 2: Caption

- How should I get my graphics?

The theory of the topic should be thoroughly discussed in this section. The student must show their proficiency on the topic as well as additional insight. This section may be separated into a few sections as *necessary*. Note that the theory of the topic is the main contribution and the report should illustrate this. In Theorem ?? we discuss..... $f(x) = \begin{cases} 1 & \text{if } x < 2 \\ 2 & \text{if } x \geq 2 \end{cases}$

Theorem 1. *fbkndfblksnfbkn*

$$jbjfbfjk bbfjf f(x) = \begin{cases} 1 & \text{if } x < 2 \\ 2 & \text{if } x \geq 2 \end{cases}$$

Proof. blksnfbldfsnb

□

m,fgkrjtbnel

Theorem 2. *aurbyveurybverybyruwei*

Proof. vnitunrivuptnriuwvtnoiutnreuvnt

□

3 Application

The application should be presented in this section. Code should be included in an appendix as well as additional output if needed.

4 Conclusion

The conclusion should summarise what was done in the research report. It should also provide shortfalls of the research and recommendations on what could be investigated in future. This section should be an honest summary of the research.

YOUR 10 PAGE COUNT ENDS HERE

klvunl ruvynyrynfsfng hdfnfgjf gjfhjghjkh jhkgjhkh dfndj ghkifgj djdj dgljhjhghj	ævirue $x = y$ n oiurlevniq	ermwovmi;q	ævpinqw	ovmi;
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Table 1: rvybruvybrv

	Column 1	Column 2
Row 1		
Row 2		

Table 2: Example

Appendix

Include any additional code, output or data here.